3.2.3 Transport across cell membranes

Content	Opportunities for skills development
The basic structure of all cell membranes, including cell-surface membranes and the membranes around the cell organelles of eukaryotes, is the same. The arrangement and any movement of phospholipids, proteins, glycoproteins and glycolipids in the fluid-mosaic model of membrane structure. Cholesterol may also be present in cell membranes where it restricts the movement of other molecules making up the membrane. Movement across membranes occurs by: • simple diffusion (involving limitations imposed by the nature of the phospholipid bilayer) • facilitated diffusion (involving the roles of carrier proteins and channel proteins) • osmosis (explained in terms of water potential) • active transport (involving the role of carrier proteins and the importance of the hydrolysis of ATP) • co-transport (illustrated by the absorption of sodium ions and glucose by cells lining the mammalian ileum). Cells may be adapted for rapid transport across their internal or external membranes by an increase in surface area of, or by an increase in the number of protein channels and carrier molecules in, their membranes. **Students should be able to:** • explain the adaptations of specialised cells in relation to the rate of transport across their internal and external membranes • explain how surface area, number of channel or carrier proteins and differences in gradients of concentration or water potential affect the rate of movement across cell membranes.	
Required practical 3: Production of a dilution series of a solute to produce a calibration curve with which to identify the water potential of plant tissue. **Required practical 4:** Investigation into the effect of a named variable on the permeability of cell-surface membranes.	**MS 3.2** Students could plot the data from their investigations in an appropriate format. **MS 3.4** Students could determine the water potential of plant tissues using the intercept of a graph of, eg, water potential of solution against gain/loss of mass.